

Equilibrium Welfare Properties

230333 Microeconomics 3 (CentER) – Part II
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Introduction

- ▶ In this section we will prove:
 - ▶ The *First Welfare Theorem*: The allocation from any competitive equilibrium with transfers is Pareto optimal.
 - ▶ The *Second Welfare Theorem*: For any Pareto optimal allocation, there is a price vector that can support it as an equilibrium with transfers.
- ▶ Both theorems require complete markets, rational and locally nonsatiated preferences, and nonempty and closed production sets.
- ▶ However, the second welfare theorem requires a number of additional assumptions.

Kenneth Joseph Arrow (1921-2017)

- ▶ Born in New York City from Romanian parents and did his PhD at Columbia under Harold Hotelling.
- ▶ Together with Gérard Debreu offered the first rigorous proof of the existence of equilibrium and the fundamental welfare theorems using topological methods.
- ▶ Won the Nobel Memorial Prize in 1972 together with John Hicks.
- ▶ Also famous for *Arrow's impossibility theorem* in social choice and the Arrow-Debreu model of stage-contingent securities (with Gérard Debreu).



Price Equilibrium with Transfers

Definition

Given an economy specified by $\left(\{(X_i, \succeq_i)\}_{i=1}^I, \{Y_j\}_{j=1}^J, \bar{\omega}\right)$, an allocation $(\mathbf{x}^\star, \mathbf{y}^\star)$ and a price vector $\mathbf{p}$ constitute a *price equilibrium with transfers* if there is an assignment of wealth levels $(w_1, \dots, w_I)$ with $\sum_{i=1}^I w_i = \mathbf{p} \cdot \bar{\omega} + \sum_{j=1}^J \mathbf{p} \cdot \mathbf{y}_j^\star$ such that

- (i) For every j , $\mathbf{y}_j^\star$ maximizes profits in Y_j ; that is,

$$\mathbf{p} \cdot \mathbf{y}_j \leq \mathbf{p} \cdot \mathbf{y}_j^\star \text{ for all } \mathbf{y}_j \in Y_j$$

- (ii) For every i , $\mathbf{x}_i^\star$ is maximal for $\succeq_i$ in the budget set:

$$\{\mathbf{x}_i \in X_i : \mathbf{p} \cdot \mathbf{x}_i \leq w_i\}$$

- (iii) $\sum_{i=1}^I \mathbf{x}_i^\star = \bar{\omega} + \sum_{j=1}^J \mathbf{y}_j^\star$.

If $w_i = \mathbf{p} \cdot \omega_i + \sum_{j=1}^J \theta_{ij} \mathbf{p} \cdot \mathbf{y}_j \forall i$, then there are no transfers.

The First Fundamental Theorem of Welfare Economics

Theorem

If preferences are locally nonsatiated, and if $(\mathbf{x}^, \mathbf{y}^*, \mathbf{p})$ is a price equilibrium with transfers, then the allocation $(\mathbf{x}^*, \mathbf{y}^*)$ is Pareto optimal.*

Proof:

1. Because $(\mathbf{x}^*, \mathbf{y}^*, \mathbf{p})$ is an equilibrium, if $\mathbf{x}_i \succ_i \mathbf{x}_i^*$, then $\mathbf{p} \cdot \mathbf{x}_i > w_i$.
2. Furthermore, if $\mathbf{x}_i \succeq_i \mathbf{x}_i^*$, then $\mathbf{p} \cdot \mathbf{x}_i \geq w_i$.
 - ▶ Suppose there is an $\mathbf{x}'_i$ satisfying $\mathbf{x}'_i \succeq_i \mathbf{x}_i^*$ but $\mathbf{p} \cdot \mathbf{x}'_i < w_i$.
 - ▶ By LNS, $\exists \mathbf{x}''_i$ arbitrarily close to $\mathbf{x}'_i$ where $\mathbf{x}''_i \succ_i \mathbf{x}'_i$ and $\mathbf{p} \cdot \mathbf{x}''_i < w_i$.
 - ▶ But this contradicts that $\mathbf{x}_i^*$ was maximal in i 's budget set, because by transitivity $\mathbf{x}''_i \succ_i \mathbf{x}_i^*$.

First Welfare Theorem Proof

3. Suppose $\exists (\mathbf{x}', \mathbf{y}')$ that Pareto dominates $(\mathbf{x}^*, \mathbf{y}^*)$.
 - ▶ By (1) & (2), $\mathbf{p} \cdot \mathbf{x}'_i \geq w_i \forall i$ and $\mathbf{p} \cdot \mathbf{x}'_i > w_i$ for at least one i .
 - ▶ So $\sum_{i=1}^I \mathbf{p} \cdot \mathbf{x}'_i > \sum_{i=1}^I w_i = \mathbf{p} \cdot \bar{\omega} + \sum_{j=1}^J \mathbf{p} \cdot \mathbf{y}^*_j$.
4. Because $\mathbf{y}^*_j$ is profit-maximizing at $\mathbf{p}$, for all j we have $\mathbf{p} \cdot \mathbf{y}^*_j \geq \mathbf{p} \cdot \mathbf{y}_j \forall \mathbf{y}_j \in Y_j$.
 - ▶ Therefore $\mathbf{p} \cdot \bar{\omega} + \sum_{j=1}^J \mathbf{p} \cdot \mathbf{y}^*_j \geq \mathbf{p} \cdot \bar{\omega} + \sum_{j=1}^J \mathbf{p} \cdot \mathbf{y}'_j$.
5. Because $(\mathbf{x}', \mathbf{y}')$ is Pareto improving: $\sum_{i=1}^I \mathbf{x}'_i = \bar{\omega} + \sum_{j=1}^J \mathbf{y}'_j$.
 - ▶ This implies $\sum_{i=1}^I \mathbf{p} \cdot \mathbf{x}'_i = \mathbf{p} \cdot \bar{\omega} + \sum_{j=1}^J \mathbf{p} \cdot \mathbf{y}'_j$
6. But (3) & (4) imply $\sum_{i=1}^I \mathbf{p} \cdot \mathbf{x}'_i > \mathbf{p} \cdot \bar{\omega} + \sum_{j=1}^J \mathbf{p} \cdot \mathbf{y}'_j$.
 - ▶ But this contradicts (5).

Separating and Supporting Hyperplane Theorems

We will use these two theorems to prove certain propositions:

Theorem (Separating Hyperplane Theorem)

Suppose that the convex sets $\mathcal{A} \subset \mathbb{R}^N$ and $\mathcal{B} \subset \mathbb{R}^N$ are disjoint. Then there is $\mathbf{p} \in \mathbb{R}^N$ with $\mathbf{p} \neq \mathbf{0}$ and a value $c \in \mathbb{R}$ such that $\mathbf{p} \cdot \mathbf{x} \geq c$ for every $\mathbf{x}$ in $\mathcal{A}$ and $\mathbf{p} \cdot \mathbf{y} \leq c$ for every $\mathbf{y} \in \mathcal{B}$.

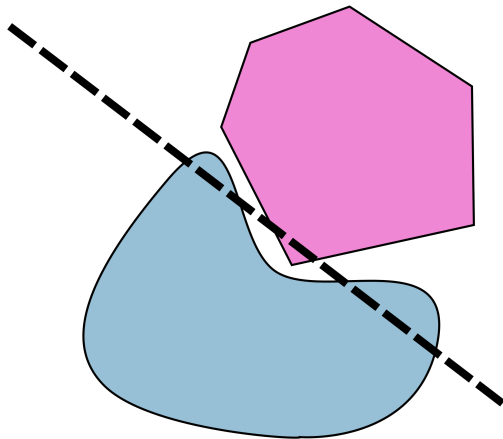
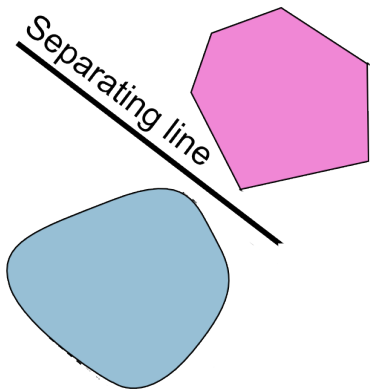
- There is a hyperplane that separates A and B , with A and B on different sides of it.

Theorem (Supporting Hyperplane Theorem)

Suppose that $\mathcal{B} \subset \mathbb{R}^N$ is convex and that $\mathbf{x}$ is not an element of the interior of the set $\mathcal{B}$. Then there is a $\mathbf{p} \in \mathbb{R}^N$ with $\mathbf{p} \neq \mathbf{0}$ such that $\mathbf{p} \cdot \mathbf{x} \geq \mathbf{p} \cdot \mathbf{y}$ for every $\mathbf{y} \in \mathcal{B}$.

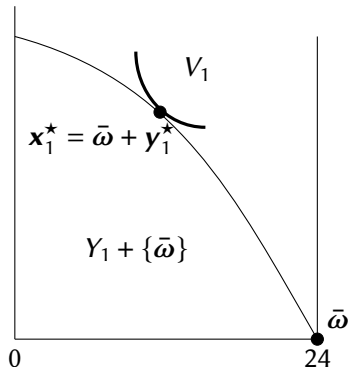
Examples

- ▶ Example 1: 2 convex, disjoint sets. SHT can be applied.
- ▶ Example 2: 2 nonconvex, disjoint sets. SHT can't be applied.



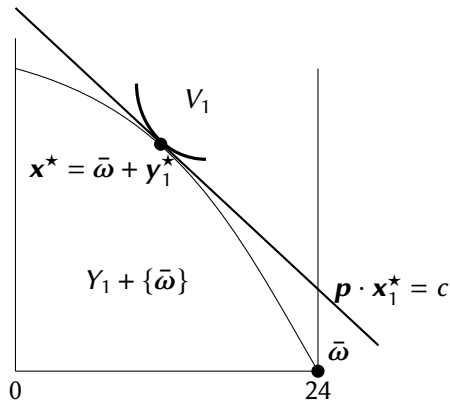
SHT Example in the Robinson Crusoe Economy

- ▶ Suppose $(\mathbf{x}_1^*, \mathbf{y}_1^*)$ is Pareto optimal.
- ▶ Crusoe's "better than set" is $V_1 = \{\mathbf{x}_1 \in X_1 : \mathbf{x}_1 \succ_1 \mathbf{x}_1^*\}$.
- ▶ The two sets V_1 and $Y_1 + \{\bar{\omega}\}$ are:
 - ▶ disjoint (by Pareto the optimality of $(\mathbf{x}_1^*, \mathbf{y}_1^*)$), and
 - ▶ convex (if $\succeq_1$ and Y_1 are convex).
- ▶ The separating hyperplane theorem can be applied.



SHT Example in the Robinson Crusoe Economy

- ▶ The SHT says $\exists \mathbf{p} \neq \mathbf{0}$ and a c such that $\mathbf{p} \cdot \mathbf{x}_1 \geq c \ \forall \mathbf{x}_1 \in V_1$ and $\mathbf{p} \cdot (\mathbf{y}_1 + \bar{\omega}) \leq c \ \forall \mathbf{y}_1 + \bar{\omega} \in Y_1 + \{\bar{\omega}\}$.



- ▶ What we will show: if we transfer wealth $w_1 = c = \mathbf{p} \cdot \mathbf{x}_1^*$ to Crusoe, $(\mathbf{x}_1^*, \mathbf{y}_1^*, \mathbf{p})$ is an equilibrium.

The Second Fundamental Theorem of Welfare Economics

Theorem

Consider an economy specified by $\left(\{(X_i, \succeq_i)\}_{i=1}^I, \{Y_j\}_{j=1}^J, \bar{\omega}\right)$, and suppose that

- ▶ *Every X_i is convex with $\mathbf{0} \in X_i$.*
- ▶ *Every preference relation $\succeq_i$ is convex, continuous and locally nonsatiated.*
- ▶ *Every Y_j is convex and exhibits free disposal.*

If $(\mathbf{x}^, \mathbf{y}^*)$ is a Pareto optimal allocation, where $\mathbf{x}_i^* \gg \mathbf{0}$ for all i , there exists a price vector $\mathbf{p} \geq \mathbf{0}$, $\mathbf{p} \neq \mathbf{0}$ such that $(\mathbf{x}^*, \mathbf{y}^*, \mathbf{p})$ is a price equilibrium with transfers.*

Thus, there is a price vector and an assignment of wealth levels $(w_1, \dots, w_I)$ satisfying $\sum_{i=1}^I w_i = \mathbf{p} \cdot \bar{\omega} + \sum_{j=1}^J \mathbf{p} \cdot \mathbf{y}_j^*$ such that $(\mathbf{x}^*, \mathbf{y}^*, \mathbf{p})$ is a Walrasian equilibrium.

Second Welfare Theorem Proof: Preliminaries

The goal is to show that the wealth levels $w_i = \mathbf{p} \cdot \mathbf{x}_i^*$ for all i support $(\mathbf{x}^*, \mathbf{y}^*, \mathbf{p})$ as a price equilibrium with transfers.

Define the sets:

- ▶ $V_i = \{\mathbf{x}_i \in X_i : \mathbf{x}_i \succ_i \mathbf{x}_i^*\} \subset \mathbb{R}^L$
- ▶ $V = \sum_{i=1}^I V_i = \{\sum_{i=1}^I \mathbf{x}_i \in \mathbb{R}^L : \mathbf{x}_1 \in V_1, \dots, \mathbf{x}_I \in V_I\}$
- ▶ $Y = \sum_{j=1}^J Y_j = \{\sum_{j=1}^J \mathbf{y}_j \in \mathbb{R}^L : \mathbf{y}_1 \in Y_1, \dots, \mathbf{y}_J \in Y_J\}$
- ▶ V is the set of aggregate consumption bundles that *could* be split across the I individuals with each i preferring it to $\mathbf{x}_i^*$.
- ▶ $Y + \{\bar{\omega}\} = \{\sum_{j=1}^J \mathbf{y}_j + \bar{\omega} \in \mathbb{R}^L : \mathbf{y}_1 \in Y_1, \dots, \mathbf{y}_J \in Y_J\}$ is the set of aggregate bundles producible with the given technology and endowments.

With this, we split the proof into multiple steps.

Second Welfare Theorem Proof Outline

Step 1 Every set V_i is convex.

Step 2 The sets V and $Y + \{\bar{\omega}\}$ are convex.

Step 3 V and $Y + \{\bar{\omega}\}$ are disjoint.

Step 4 There is a vector $\mathbf{p} \geq \mathbf{0}$, $\mathbf{p} \neq \mathbf{0}$ and a number c such that $\mathbf{p} \cdot \mathbf{z} \geq c$ for every $\mathbf{z} \in V$ and $\mathbf{p} \cdot \mathbf{z} \leq c$ for every $\mathbf{z} \in Y + \{\bar{\omega}\}$.

Step 5 If $\mathbf{x}_i \succeq_i \mathbf{x}_i^*$ for every i , then $\mathbf{p} \cdot \left(\sum_{i=1}^I \mathbf{x}_i \right) \geq c$.

Step 6 $\mathbf{p} \cdot \left(\sum_{i=1}^I \mathbf{x}_i^* \right) = \mathbf{p} \cdot \left(\bar{\omega} + \sum_{j=1}^J \mathbf{y}_j^* \right) = c$.

Step 7 For every j , we have $\mathbf{p} \cdot \mathbf{y}_j \leq \mathbf{p} \cdot \mathbf{y}_j^*$ for all $\mathbf{y}_j \in Y_j$.

Step 8 For every i , if $\mathbf{x}_i \succ_i \mathbf{x}_i^*$, then $\mathbf{p} \cdot \mathbf{x}_i > \mathbf{p} \cdot \mathbf{x}_i^*$.

Step 9 Steps 7 & 8 with feasibility from the Pareto optimal allocation implies that the wealth levels $w_i = \mathbf{p} \cdot \mathbf{x}_i^*$ for all i support $(\mathbf{x}^*, \mathbf{y}^*, \mathbf{p})$ as a price equilibrium with transfers.

Second Welfare Theorem Proof

Step 1

Every set $V_i = \{\mathbf{x}_i \in X_i : \mathbf{x}_i \succsim_i \mathbf{x}_i^*\}$ is convex.

- ▶ We need to show that if $\mathbf{x}_i \in V_i$ and $\mathbf{x}'_i \in V_i$, then $\mathbf{x}_i^\alpha = \alpha\mathbf{x}_i + (1 - \alpha)\mathbf{x}'_i \in V_i$ for all $\alpha \in [0, 1]$.
- ▶ First, by the convexity of X_i , $\mathbf{x}_i^\alpha \in X_i$.
- ▶ $\mathbf{x}_i, \mathbf{x}'_i \in V_i$ means $\mathbf{x}_i \succsim_i \mathbf{x}_i^*$ and $\mathbf{x}'_i \succsim_i \mathbf{x}_i^*$.
- ▶ Suppose wlog that $\mathbf{x}_i \succeq_i \mathbf{x}'_i$.
- ▶ Because preferences are convex: $\mathbf{x}_i^\alpha \succeq_i \mathbf{x}'_i \forall \alpha \in [0, 1]$
- ▶ Then by transitivity $\mathbf{x}_i^\alpha \succsim_i \mathbf{x}_i^*$.
- ▶ Hence $\mathbf{x}_i^\alpha \in V_i$.

Second Welfare Theorem Proof

Step 2

The sets V and $Y + \{\bar{\omega}\}$ are convex.

- ▶ The sum of convex sets is convex (proof on next slide for $V_1 + V_2$ case).

If V_1 and V_2 are convex, $V = V_1 + V_2$ is convex

- ▶ Take $\mathbf{x}' = \mathbf{x}'_1 + \mathbf{x}'_2 \in V$ and $\mathbf{x}'' = \mathbf{x}''_1 + \mathbf{x}''_2 \in V$.
- ▶ WTS: $\forall \alpha \in [0, 1]$ that $\alpha \mathbf{x}' + (1 - \alpha) \mathbf{x}'' \in V$.
- ▶ $\mathbf{x}'_1 \in V_1$ and $\mathbf{x}''_1 \in V_1$ and similarly for $\mathbf{x}'_2$ and $\mathbf{x}''_2$.
- ▶ Because V_1 and V_2 are convex, $\forall \alpha \in [0, 1]$, $\mathbf{x}_1^\alpha = \alpha \mathbf{x}'_1 + (1 - \alpha) \mathbf{x}''_1 \in V_1$ and similarly $\mathbf{x}_2^\alpha \in V_2$.
- ▶ So, by the definition of V :

$$\begin{aligned}\alpha \mathbf{x}' + (1 - \alpha) \mathbf{x}'' &= \alpha (\mathbf{x}'_1 + \mathbf{x}'_2) + (1 - \alpha) (\mathbf{x}''_1 + \mathbf{x}''_2) \\ &= \alpha \mathbf{x}'_1 + (1 - \alpha) \mathbf{x}''_1 + \alpha \mathbf{x}'_2 + (1 - \alpha) \mathbf{x}''_2 \\ &= \mathbf{x}_1^\alpha + \mathbf{x}_2^\alpha\end{aligned}$$

- ▶ This is an element of V since it is the sum of two vectors which are each elements of V_1 and V_2 .

Second Welfare Theorem Proof

Step 3

V and $Y + \{\bar{\omega}\}$ are disjoint.

- ▶ V contains all bundles that can be distributed such that everyone is strictly better off than with $\mathbf{x}_i^*$.
- ▶ $Y + \{\bar{\omega}\}$ is the set of all feasible bundles.
- ▶ If they were not disjoint, then $(\mathbf{x}^*, \mathbf{y}^*)$ would not be Pareto optimal.

Second Welfare Theorem Proof

Step 4

There is a vector $\mathbf{p} \geq \mathbf{0}$, $\mathbf{p} \neq \mathbf{0}$ and a number c such that $\mathbf{p} \cdot \mathbf{z} \geq c$ for every $\mathbf{z} \in V$ and $\mathbf{p} \cdot \mathbf{z} \leq c$ for every $\mathbf{z} \in Y + \{\bar{\omega}\}$.

- ▶ That such a $\mathbf{p} \in \mathbb{R}^L$, $\mathbf{p} \neq \mathbf{0}$ exists follows directly from the separating hyperplane theorem (two disjoint convex sets).
- ▶ We only need to rule out the possibility of $p_\ell < 0$ for any ℓ .
- ▶ Because firms have free disposal, if $p_\ell < 0$ then $\mathbf{p} \cdot \mathbf{y}_j$ could become unboundedly large, violating $\mathbf{p} \cdot \mathbf{z} \leq c$ for all $\mathbf{z} \in Y + \{\bar{\omega}\}$.

Second Welfare Theorem Proof

Step 5

If $\mathbf{x}_i \succeq_i \mathbf{x}_i^*$ for every i , then $\mathbf{p} \cdot \left(\sum_{i=1}^I \mathbf{x}_i \right) \geq c$.

- ▶ Take $\mathbf{x}_i \succeq_i \mathbf{x}_i^*$. By LNS we have, $\forall \varepsilon > 0$, $\exists \hat{\mathbf{x}}_i$ satisfying $\|\hat{\mathbf{x}}_i - \mathbf{x}_i\| \leq \varepsilon$ such that $\hat{\mathbf{x}}_i \succ_i \mathbf{x}_i$.
- ▶ By transitivity $\hat{\mathbf{x}}_i \succ_i \mathbf{x}_i^*$ so $\hat{\mathbf{x}}_i \in V_i$.
- ▶ Such a $\hat{\mathbf{x}}_i$ exists for every consumer, so $\sum_{i=1}^I \hat{\mathbf{x}}_i \in V$.
- ▶ By Step 4: $\mathbf{p} \cdot \left(\sum_{i=1}^I \hat{\mathbf{x}}_i \right) \geq c$.
- ▶ As $\varepsilon \rightarrow 0$ (so $\hat{\mathbf{x}}_i \rightarrow \mathbf{x}_i \forall i$), we have $\mathbf{p} \cdot \left(\sum_{i=1}^I \mathbf{x}_i \right) \geq c$.
 - ▶ Limits preserve inequalities (see proof on next slide).

Note: Limits Preserve Inequalities

- ▶ Consider the sequence $\sum_{i=1}^l \widehat{\mathbf{x}}_i \rightarrow \sum_{i=1}^l \mathbf{x}_i$ where $\mathbf{p} \cdot \left(\sum_{i=1}^l \widehat{\mathbf{x}}_i \right) \geq c$.
- ▶ We want to show that this inequality is preserved at the limit: $\mathbf{p} \cdot \left(\sum_{i=1}^l \mathbf{x}_i \right) \geq c$.
- ▶ Suppose toward a contradiction that instead $\mathbf{p} \cdot \left(\sum_{i=1}^l \mathbf{x}_i \right) = d < c$.
- ▶ From the definition of the limit of a function:

$$\lim_{\sum_{i=1}^l \widehat{\mathbf{x}}_i \rightarrow \sum_{i=1}^l \mathbf{x}_i} \mathbf{p} \cdot \left(\sum_{i=1}^l \widehat{\mathbf{x}}_i \right) = d$$

implies that $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall \sum_{i=1}^l \widehat{\mathbf{x}}_i, 0 < \left| \sum_{i=1}^l \widehat{\mathbf{x}}_i - \sum_{i=1}^l \mathbf{x}_i \right| < \delta$ implies that $\left| \mathbf{p} \cdot \left(\sum_{i=1}^l \widehat{\mathbf{x}}_i \right) - d \right| < \varepsilon$.

- ▶ This holds for all $\varepsilon > 0$. Choose $\varepsilon = c - d$. $\exists \delta > 0$ s.t. $\forall \sum_{i=1}^l \widehat{\mathbf{x}}_i,$
 $0 < \left| \sum_{i=1}^l \widehat{\mathbf{x}}_i - \sum_{i=1}^l \mathbf{x}_i \right| < \delta \implies \left| \mathbf{p} \cdot \left(\sum_{i=1}^l \widehat{\mathbf{x}}_i \right) - d \right| < \varepsilon = c - d$.
- ▶ But then:

$$-\varepsilon < \mathbf{p} \cdot \left(\sum_{i=1}^l \widehat{\mathbf{x}}_i \right) - d < \varepsilon = c - d \implies \mathbf{p} \cdot \left(\sum_{i=1}^l \widehat{\mathbf{x}}_i \right) < c \implies \text{Contradiction!}$$

Second Welfare Theorem Proof

Step 6

$$\mathbf{p} \cdot \left(\sum_{i=1}^I \mathbf{x}_i^* \right) = \mathbf{p} \cdot \left(\bar{\omega} + \sum_{j=1}^J \mathbf{y}_j^* \right) = c.$$

- ▶ By feasibility, $\sum_{i=1}^I \mathbf{x}_i^* = \sum_{j=1}^J \mathbf{y}_j^* + \bar{\omega} \in Y + \{\bar{\omega}\}$.
- ▶ Therefore $\mathbf{p} \cdot \left(\sum_{i=1}^I \mathbf{x}_i^* \right) \leq c$ because $\mathbf{p} \cdot \mathbf{z} \leq c$ for every $\mathbf{z} \in Y + \{\bar{\omega}\}$.
- ▶ But Step 5 implies that $\mathbf{p} \cdot \left(\sum_{i=1}^I \mathbf{x}_i^* \right) \geq c$
- ▶ Therefore $\mathbf{p} \cdot \left(\sum_{i=1}^I \mathbf{x}_i^* \right) = c$.

Second Welfare Theorem Proof

Step 7

For every j , we have $\mathbf{p} \cdot \mathbf{y}_j \leq \mathbf{p} \cdot \mathbf{y}_j^\star$ for all $\mathbf{y}_j \in Y_j$.

- ▶ For all firms, $\forall \mathbf{y}_j \in Y_j$ we have $\mathbf{y}_j + \sum_{h \neq j} \mathbf{y}_h^\star \in Y$.
- ▶ From Steps 4 and 6, $\forall \mathbf{y}_j \in Y_j$:

$$\mathbf{p} \cdot \left(\bar{\omega} + \mathbf{y}_j + \sum_{h \neq j} \mathbf{y}_h^\star \right) \leq c = \mathbf{p} \cdot \left(\bar{\omega} + \mathbf{y}_j^\star + \sum_{h \neq j} \mathbf{y}_h^\star \right)$$

- ▶ Cancelling terms yields $\mathbf{p} \cdot \mathbf{y}_j \leq \mathbf{p} \cdot \mathbf{y}_j^\star$ for all $\mathbf{y}_j \in Y_j$, for all j .

Second Welfare Theorem Proof

Step 8

For every i , if $\mathbf{x}_i \succ_i \mathbf{x}_i^\star$, then $\mathbf{p} \cdot \mathbf{x}_i > \mathbf{p} \cdot \mathbf{x}_i^\star$.

- ▶ If $\mathbf{x}_i \succ_i \mathbf{x}_i^\star$, then $\mathbf{x}_i \in V_i$. From Steps 5 and 6 above we have:

$$\mathbf{p} \cdot \left(\mathbf{x}_i + \sum_{k \neq i} \mathbf{x}_k^\star \right) \geq c = \mathbf{p} \cdot \left(\mathbf{x}_i^\star + \sum_{k \neq i} \mathbf{x}_k^\star \right)$$

- ▶ Cancelling terms yields $\mathbf{p} \cdot \mathbf{x}_i \geq \mathbf{p} \cdot \mathbf{x}_i^\star$.
- ▶ Now we just need to rule out the $\mathbf{p} \cdot \mathbf{x}_i = \mathbf{p} \cdot \mathbf{x}_i^\star$ case.

Second Welfare Theorem Proof

- ▶ Suppose toward a contradiction there is a $\mathbf{x}'_i \in \mathbb{R}_+^L$ satisfying $\mathbf{x}'_i \succ_i \mathbf{x}_i^*$ such that $\mathbf{p} \cdot \mathbf{x}'_i = \mathbf{p} \cdot \mathbf{x}_i^*$.
- ▶ Because $\mathbf{0} \in X_i$ and X_i is convex, $\alpha \mathbf{x}'_i + (1 - \alpha) \mathbf{0} \in X_i$ for all $\alpha \in [0, 1]$.
- ▶ Because $\mathbf{p} \geq \mathbf{0}$, $\mathbf{p} \neq \mathbf{0}$ and $\mathbf{x}_i^* \gg \mathbf{0}$, we know that $\mathbf{p} \cdot \mathbf{x}_i^* > 0$.
- ▶ $\forall \alpha \in [0, 1)$, $\alpha \mathbf{p} \cdot \mathbf{x}'_i + (1 - \alpha) \mathbf{p} \cdot \mathbf{0} < \mathbf{p} \cdot \mathbf{x}_i^*$.
- ▶ By continuity, for α close enough to 1, $\alpha \mathbf{x}'_i \succ_i \mathbf{x}_i^*$ (see note for full proof).
- ▶ As we have found a bundle that is preferred to $\mathbf{x}_i^*$ and is strictly cheaper, we have found a contradiction to what we found above.

Second Welfare Theorem Proof

Step 9

If we assign wealth levels $w_i = \mathbf{p} \cdot \mathbf{x}_i^*$ to each consumer, $(\mathbf{x}^*, \mathbf{y}^*, \mathbf{p})$ is a price equilibrium with transfers.

This satisfies all the conditions for equilibrium:

- ▶ By Step 8: If $\mathbf{x}_i \succ_i \mathbf{x}_i^*$, then $\mathbf{p} \cdot \mathbf{x}_i > w_i, \forall i$.
 - ▶ $\mathbf{x}_i^*$ is maximal for $\succeq_i$ in the budget set.
- ▶ By Step 7: $\mathbf{p} \cdot \mathbf{y}_j \leq \mathbf{p} \cdot \mathbf{y}_j^*$ for all $\mathbf{y}_j \in Y_j, \forall j$
 - ▶ $\mathbf{y}_j^*$ maximizes profits in Y_j .
- ▶ Because $(\mathbf{x}^*, \mathbf{y}^*)$ is Pareto optimal, we have feasibility and hence market clearing in each good:

$$\sum_{i=1}^I \mathbf{x}_i^* = \bar{\omega} + \sum_{j=1}^J \mathbf{y}_j^*$$